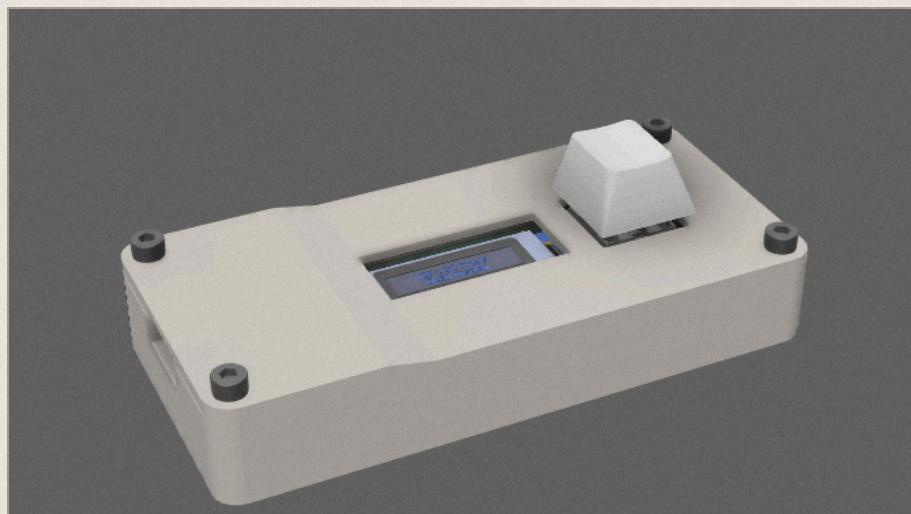


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MICRO PALETTE



Cubic-Vyper • 6 journal entries • Started Mar 18, 2026

this device will detect the color of something and display with a led, and an oled display

EARNED BADGES

I2C

Displays

Digital Sensors

[View on GitHub](#)

JOURNAL



Cubic-Vyper added to the journal **DESIGN** 0.42h logged

27d ago

so I just found out that alibaba charges aditinal shipping fees (20+), so now I need to find a new suplier to buy from

Oled (with free shipping!):



0.91 Inch SSD1306 OLED Display Module I2C 128x32 for Arduino ESP32 STM32 Raspberry Pi Blue White Color Optional Pre-soldered

★★★★★ 3.0 1 Review | 31 sold

OUTDOOR FUN Ends: Apr 7, 23:59 (GMT-7)

\$1.99 Save \$6.36
\$8.95

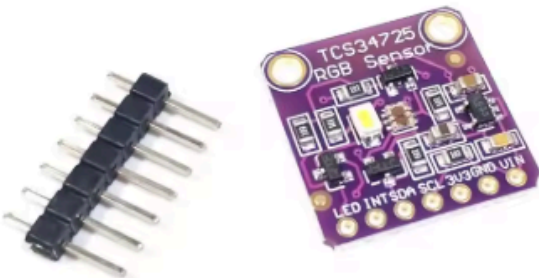
One-time offer, applies to one item only
Incl. import charges; est. other taxes: \$0.34 ⓘ ; Extra 1% off with coins

🔖 \$2.00 off on \$15.00 >

Color: 1 pcs white 0.91



Now I need a TCS34725 color sensor (with free shipping!)



RGB Color Sensor TCS34725 with IR Filter and White LED for Arduino UNO R3

21 sold

\$3.84

Incl. import charges; est. other taxes: \$0.32 ⓘ ; Extra 2% off with coins

🔖 \$2.00 off on \$15.00 >

Product sellpoints

- Optimized for Arduino UNO R3: Designed for seamless integration with Arduino UNO R3, this module is a perfect fit for hobbyists and professionals alike.

Tcs34725 Arduino High Sensitivity & Wide Dynamic Range TCS34725 Color

Now I need a Seeed Studio XIAO RP2040, this was harder than I thought, I was looking on aliexpress, eventaly I found that it was cheaper on the seedstudio website



Seeed Studio XIAO RP2040 - Supports Arduino, MicroPython and CircuitPython

[Request Custom/ Bulk Quote >](#)

USD **\$3.99**

10+: \$3.50

Ships from California

Duty-free. Sales tax & shipping applied at checkout.

XIAO Dev Board : XIAO RP2040

XIAO SAMD21

XIAO RP2040

XIAO nRF52840

XIAO nRF52840 Plus

XIAO nRF52840 Sense

and only \$5.47 for shipping!



Cubic-Vyper added to the journal **DESIGN** 0.5h logged

Mar 22, 2026

Now I fix the code so that the scan will happen only when the button is pressed

I added the pin definition:

```
#SK6812 pin
LED_DATA = board.D5

#Switch pins
SWITCH_PIN = board.d6

#start sensor i2c
sensor_i2c = busio.I2C(SENSOR_SCL, SENSOR_SDA)
```

then the switch setup:

```
#Switch setup
switch = digitalio.DigitalInOut(SWITCH_PIN)
switch.direction = digitalio.Direction.INPUT
switch.pull = digitalio.Pull.UP
```

now i also changed this message from scanning to press button:

```
update_oled("Press Button...", 0, 0, 0)
```

updating the main loop

I set the leds off on startup:

```
#turn off LEDs
Pixels.fill((0, 0, 0))
```

now I update the rest of the main loop:

```
while True:
    try:
        if not switch.value: # button pressed

            #values
            r16, g16, b16 = sensor.color_raw

            #convert to 8-bit
            r8, g8, b8 = convert_to_8bit(r16, g16, b16)

            #find closest
            color_name = find_closest_color(r8, g8, b8)

            #update oled
            update_oled(color_name, r16, g16, b16)

            #update leds
            update_leds(r8, g8, b8)

            #prevent mutiple scans in one press
            time.sleep(0.2)

            while not switch.value:
                time.sleep(0.05)

            time.sleep(0.1)

            update_oled("Press Button...", 0, 0, 0)

        else:
            time.sleep(0.05)

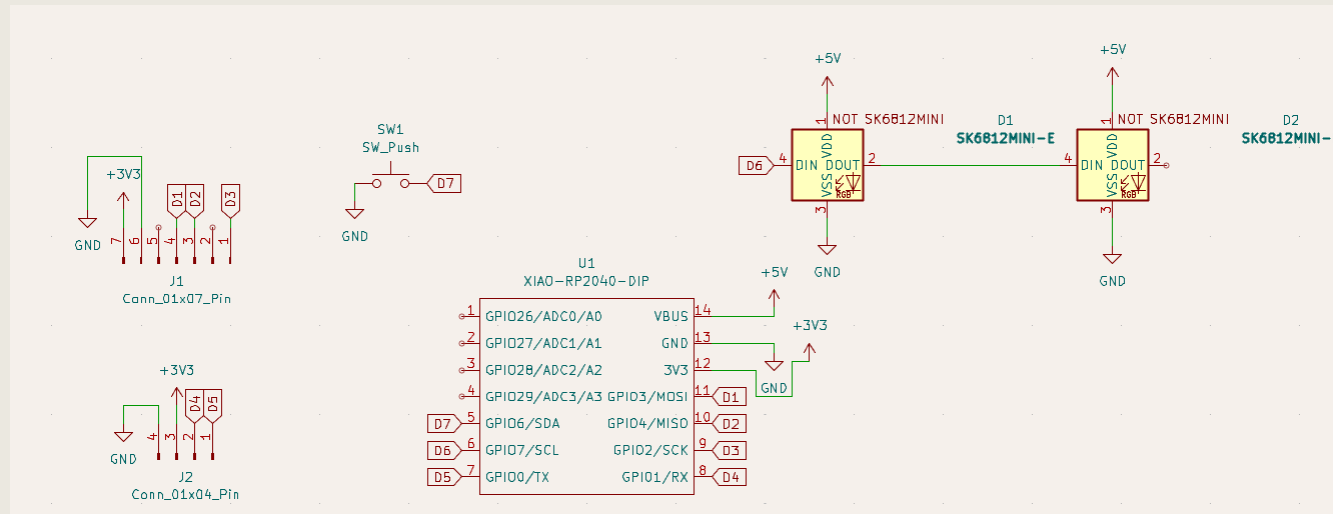
    except Exception as e:
        #show error
        print("Error:", e)
        update_oled("Error", 0, 0, 0)
```



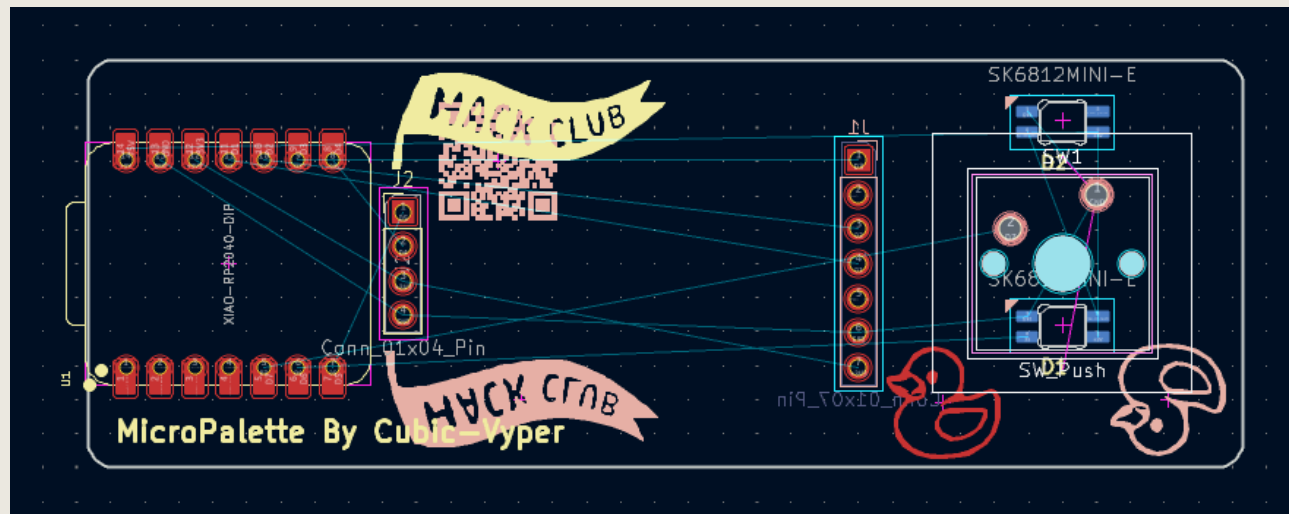
Cubic-Vyper added to the journal **DESIGN** 1.5h logged

Mar 21, 2026

So, I just realized I don't have a button to activate it, so now I will add a keyboard switch to go over the leds:

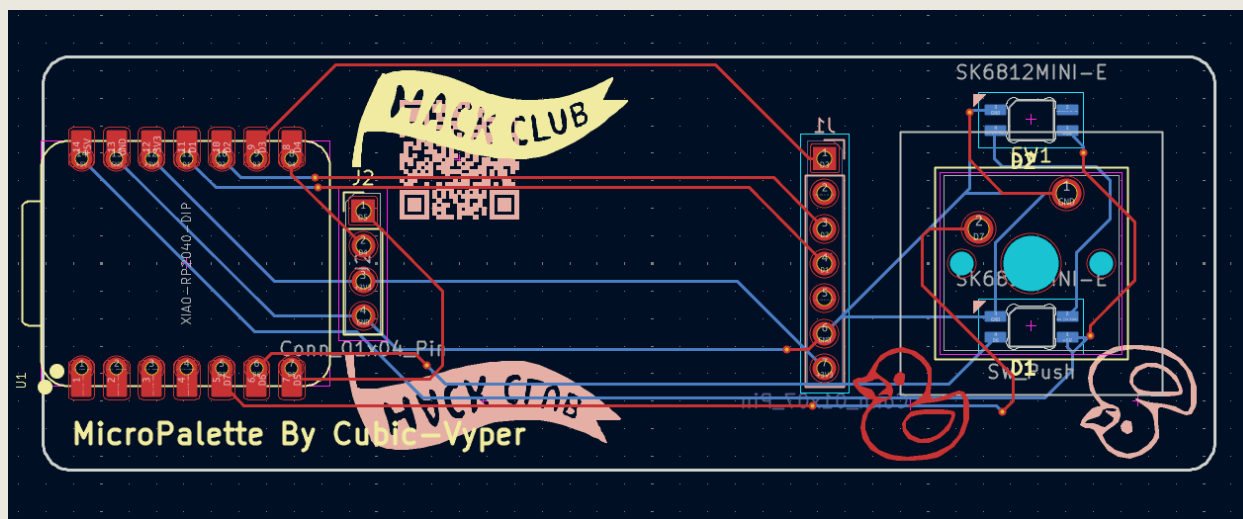


now I fix the PCB:

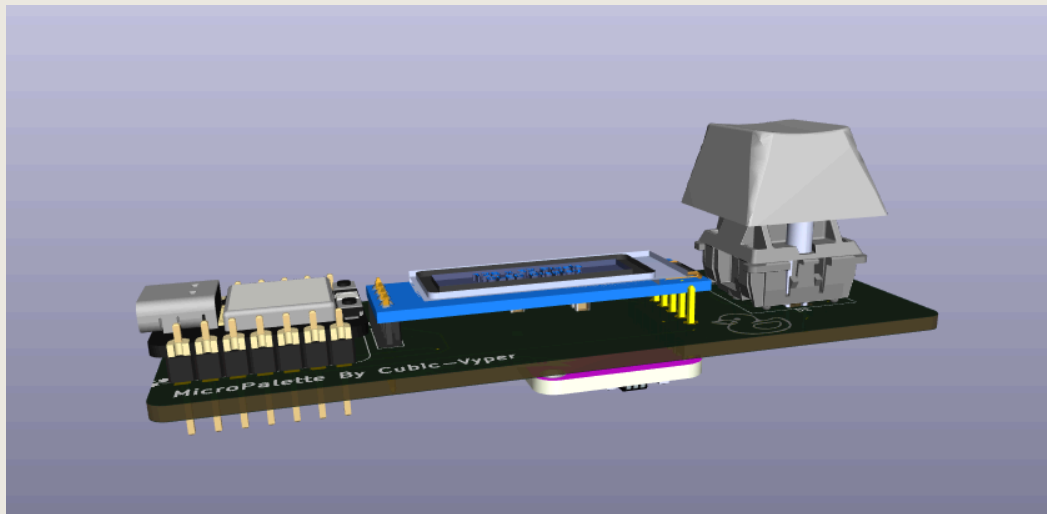


I also had to move the graphics around

now I re-wire it:



i now add the 3d model:



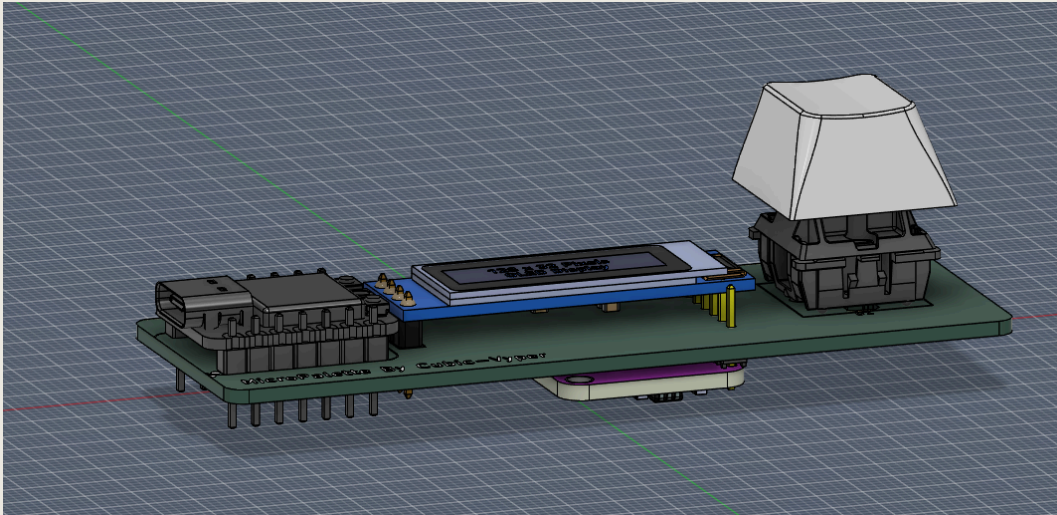
as I was trying to add this to fusion, it crashed every time:



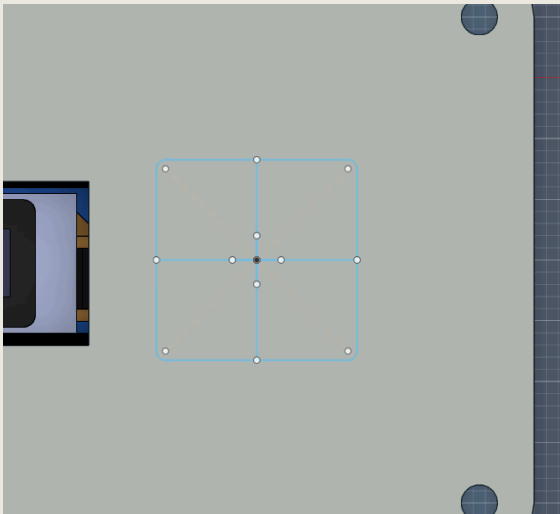
Fusion Error Report

A software problem has caused Fusion to close unexpectedly.

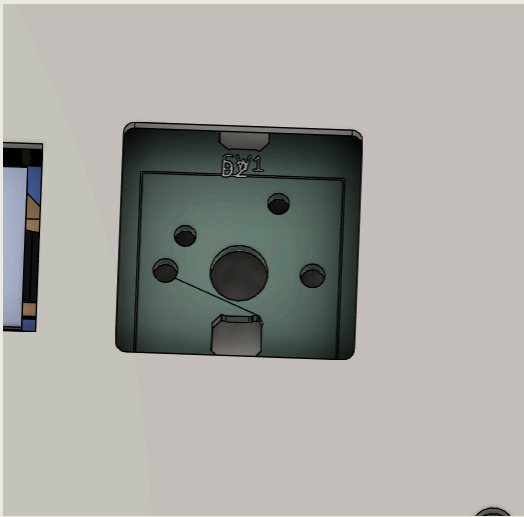
so far 5 times, it happens when I try to make the pcb solid insted of a suface, so it can stay a surface :) (fiy fusion takes a few minutes to start):



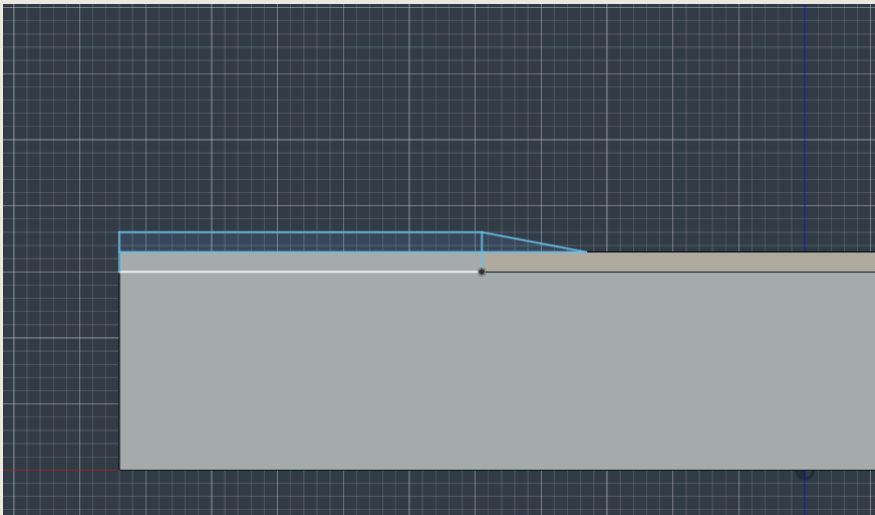
now I sketch where the hole for the switch will be:

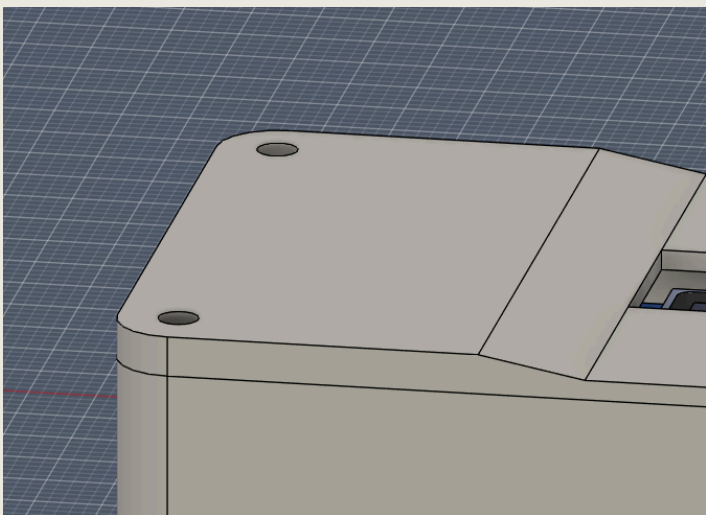
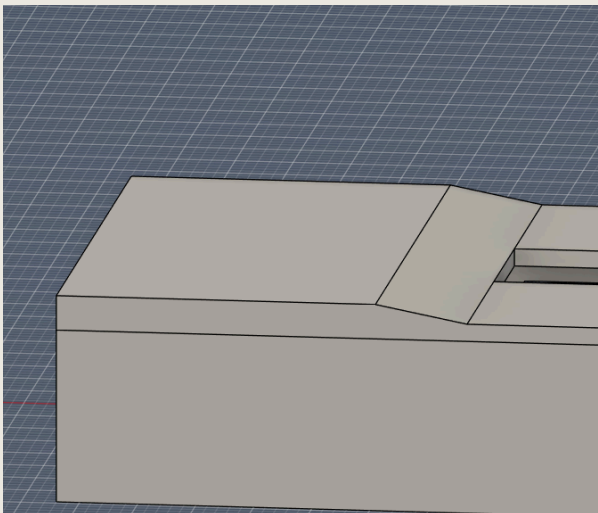


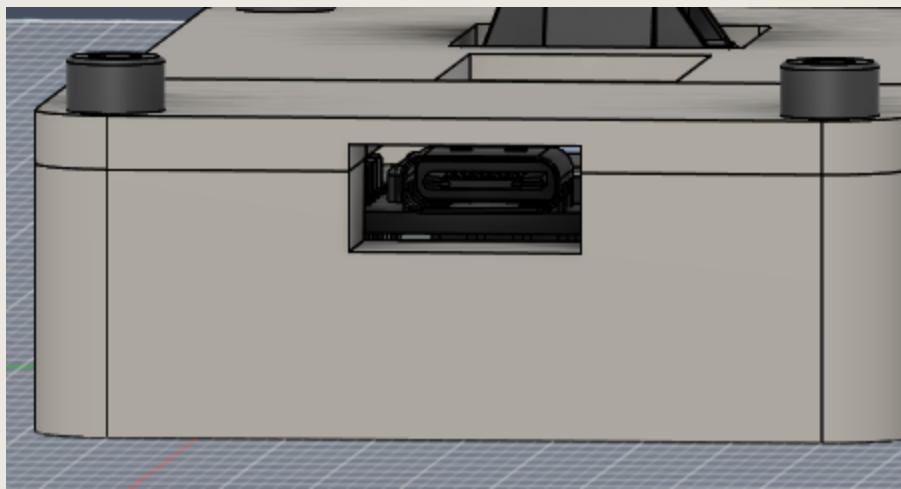
and extrude:



turns out I didn't make enough room for the xioa, and the usb-c cord, so I will expand it and make a slant so it won't be exposed:







Cubic-Vyper added to the journal **DESIGN** 2h logged

Mar 20, 2026

I forgot to setup the timlapse so I will just explain my final code

first I import libarys:

```
import board
import busio
import time
import neopixel
import adafruit_tcs34725
import adafruit_ssd1306
import digitalio
```

Then define pins:

```
#define pins

# Sensor I2C pins
SENSOR_SCL = board.D0
SENSOR_SDA = board.D1

#sensor led (illuminates object in dark case)
SENSOR_LED = board.D2

# oled I2C pins
OLED_SCL = board.D3
OLED_SDA = board.D4

#SK6812 pin
LED_DATA = board.D5
```

Then I start I2c communication and turn on sensor:

```
#start sensor i2c
sensor_i2c = busio.I2C(SENSOR_SCL, SENSOR_SDA)

#start color sensor
sensor = adafruit_tcs34725.TCS34725(sensor_i2c)
sensor.integration_time = 50 #sensor collection time
sensor.gain = 4 #amplify signal
```

Then I turn on the sensors led to light up the object (it will be in a dark case):

```
#turn on sensor LED
sensor_led = digitalio.DigitalInOut(SENSOR_LED)
sensor_led.direction = digitalio.Direction.OUTPUT
sensor_led.value = True
print("sensor LED on")
```

Then I start oled I2c, and start up the oled and leds:

```
#start oled i2c
oled_i2c = busio.I2C(OLED_SCL, OLED_SDA)

#start oled display
oled = adafruit_ssd1306.SSD1306_I2C(128, 32, oled_i2c)

#start SK6812 LEDs
Pixels = neopixel.NeoPixel(LED_DATA, 2, brightness=0.3, auto_write=True)
```

Next I define the known color codes(for oled to display):

```
#color codes
COLORS = {
    "red": (255, 0, 0),
    "green": (0, 255, 0),
    "blue": (0, 0, 255),
    "yellow": (255, 255, 0),
    "cyan": (0, 255, 255),
    "magenta": (255, 0, 255),
    "white": (255, 255, 255),
    "black": (0, 0, 0),
    "orange": (255, 165, 0),
    "purple": (128, 0, 128)
}
```

This function converts the sensor's 16bit to 8bit:

```
def convert_to_8bit(r, g, b):
    r8 = r * 255 // 65535
    g8 = g * 255 // 65535
    b8 = b * 255 // 65535
    return r8, g8, b8
```

This function finds the closest know color to display on the oled, but shows unkwn if unidentifiable:

```
def find_closest_color(r8, g8, b8):
    closest_name = "unknown"
    smallest_distance = 999999

    #compare color
    for name, (cr, cg, cb) in COLORS.items():
        #calculate color difference
        distance = (r8 - cr) ** 2 + (g8 - cg) ** 2 + (b8 - cb) ** 2

        if distance < smallest_distance:
            smallest_distance = distance
            closest_name = name

    #only return color name if it's close enough
    if smallest_distance > 20000:
        return "unknown"
    else:
        return closest_name
```

This function displays the closest color name and the values(from scanned color - 16bit for more acuracy):

```
def update_oled(color_name, r16, g16, b16):
    #clear display
    oled.fill(0)

    #show color name
    oled.text(color_name, 0, 0, 1)

    #show raw 16-bit RG values
    oled.text(f"R:{r16:4d} G:{g16:4d}", 0, 12, 1)

    #show raw 16-bit blue value
    oled.text(f"B:{b16:4d}", 0, 22, 1)

    #update display
    oled.show()
```

This updates the led with scanned color(in 8bit):

```
def update_leds(r8, g8, b8):
    Pixels.fill((r8, g8, b8))
```

finally, the main loop this scans the object and updates everything (and will show an error in needed)

```
#main loop
update_oled("scanning...", 0, 0, 0)

while True:
    try:
        #read 16-bit values
        r16, g16, b16 = sensor.color_raw

        #convert to 8-bit
        r8, g8, b8 = convert_to_8bit(r16, g16, b16)

        #find closest color
        color_name = find_closest_color(r8, g8, b8)

        #update oled with color name and 16-bit values
        update_oled(color_name, r16, g16, b16)

        #update leds
        update_leds(r8, g8, b8)

        #0.5 second delay before next scan -- this will prevent accidental scans
        time.sleep(0.5)

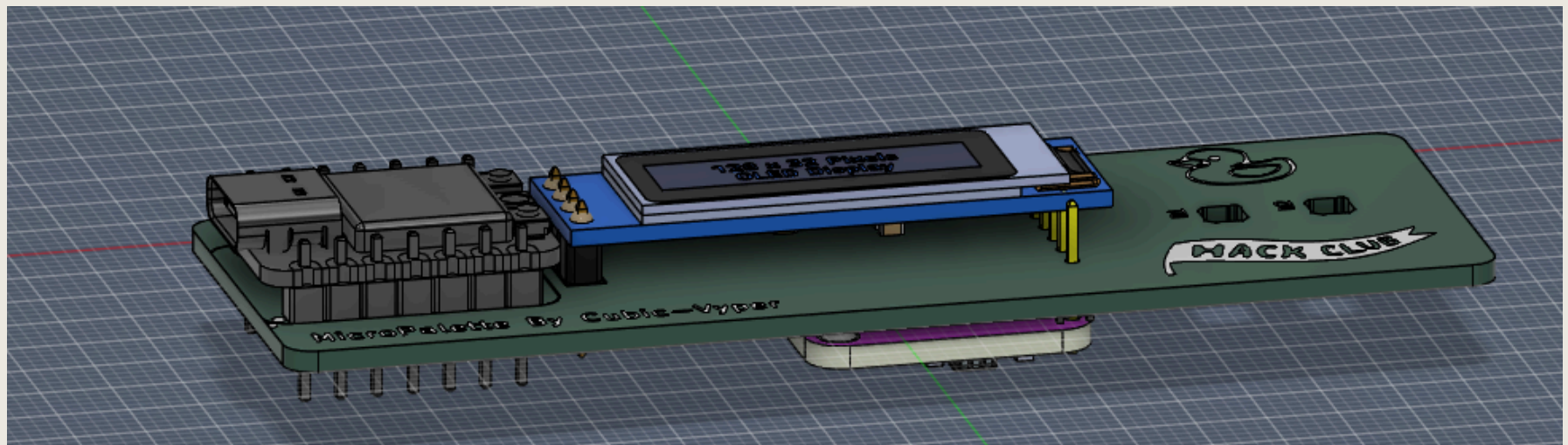
    except Exception as e:
        #show error
        print("Error:", e)
        update_oled("Error", 0, 0, 0)
        time.sleep(1)
```



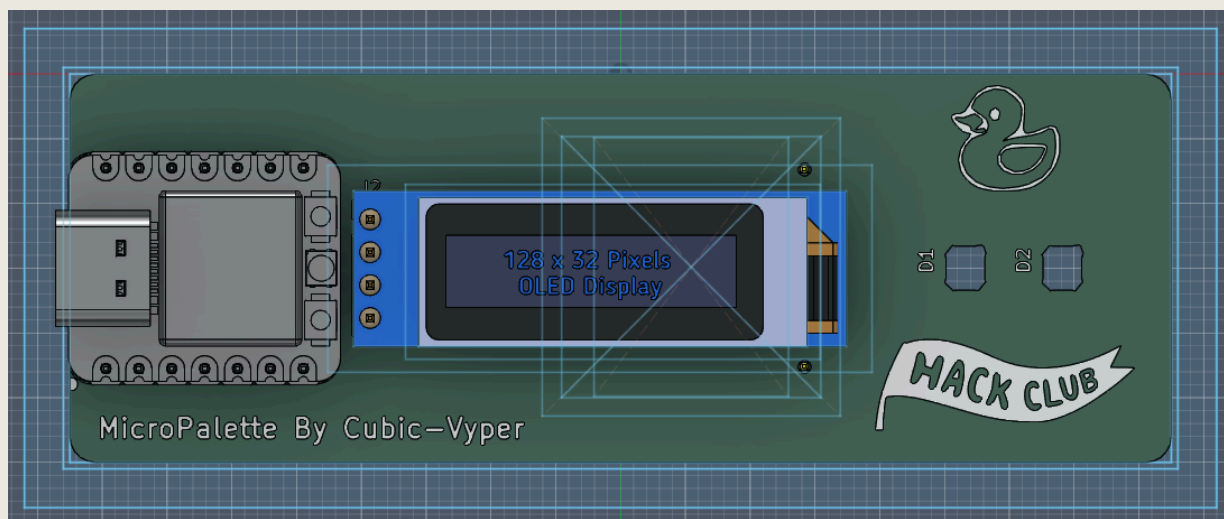
Cubic-Vyper added to the journal **DESIGN** 2.25h logged

Mar 19, 2026

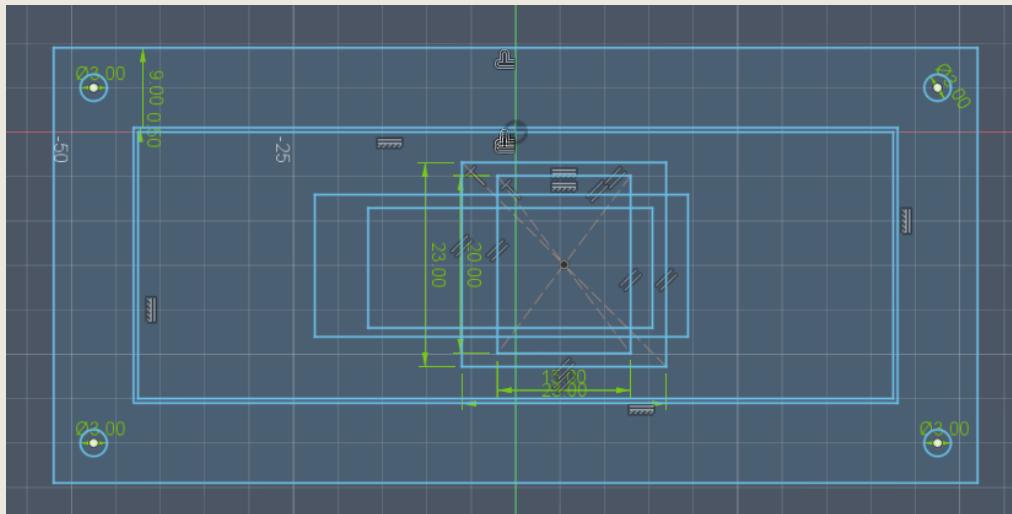
First I import the pcb with the components:



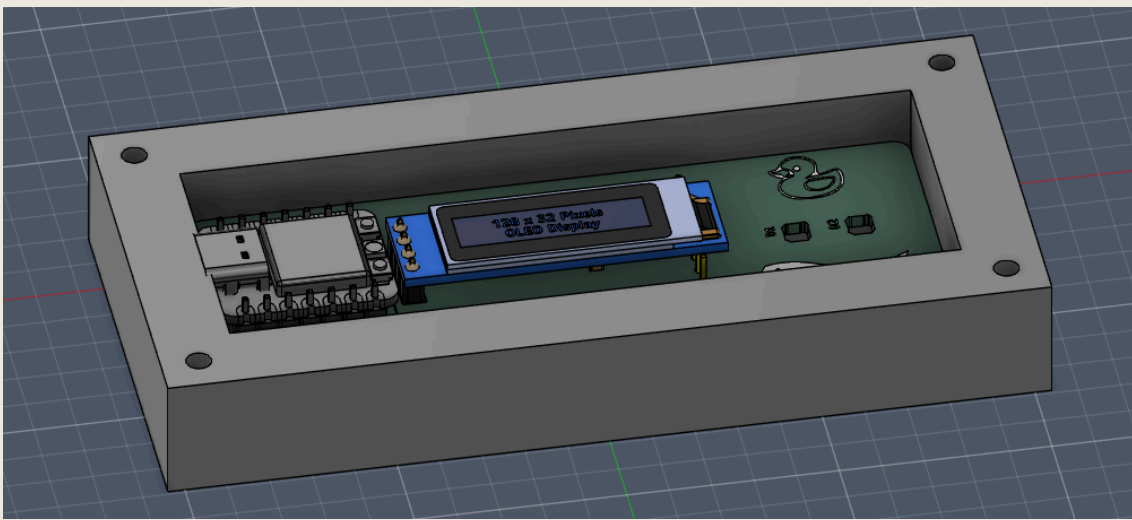
then I make a sketch with areas for all of the components:



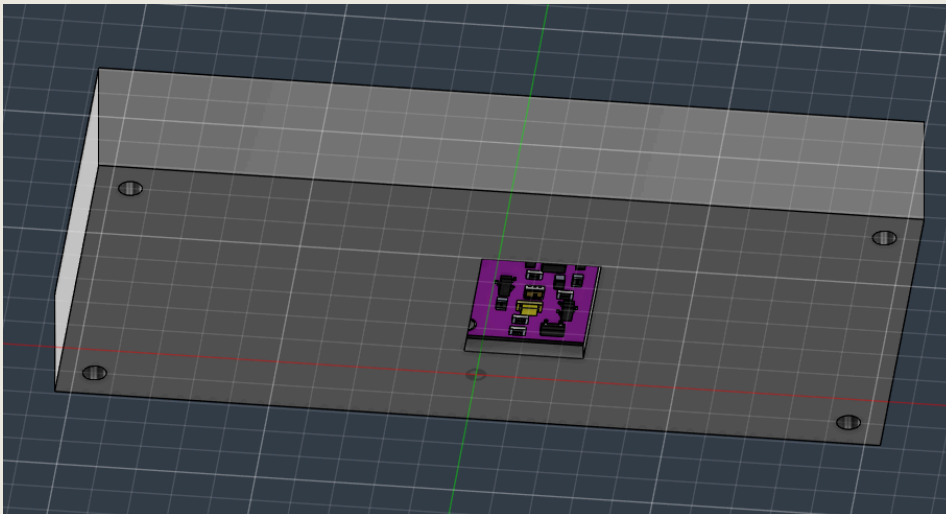
forgot to add the holes for screws:



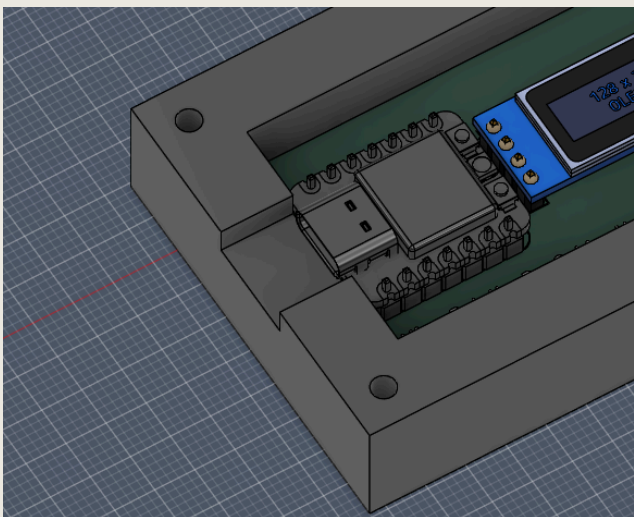
now I extrude the walls and bottem:



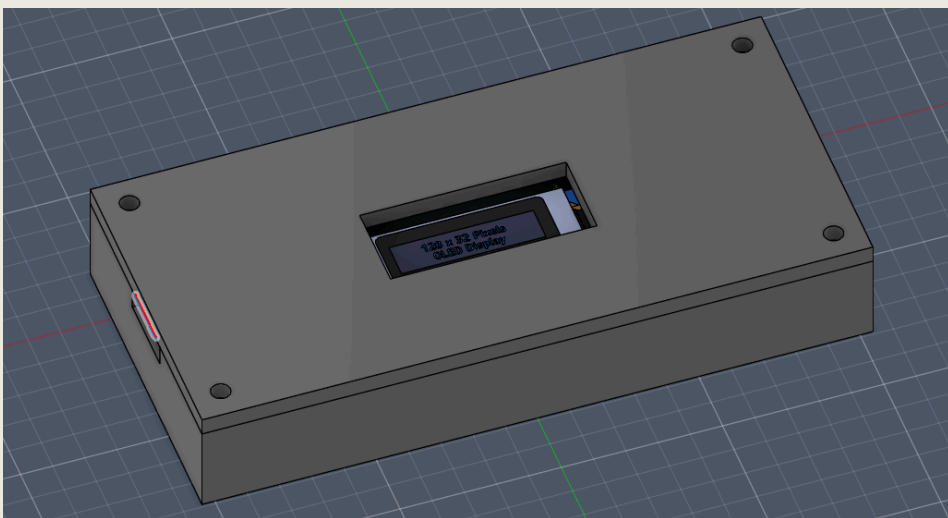
with the hole for the sensor:



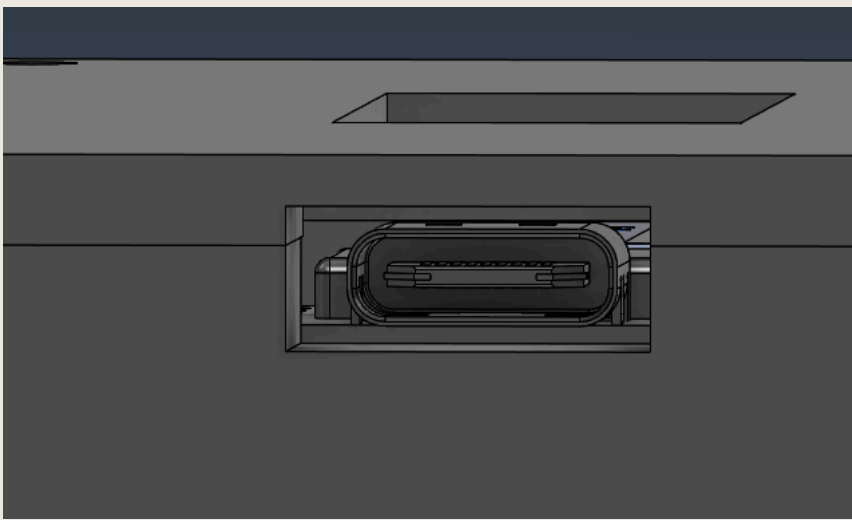
now I make a cutout for the usb c:



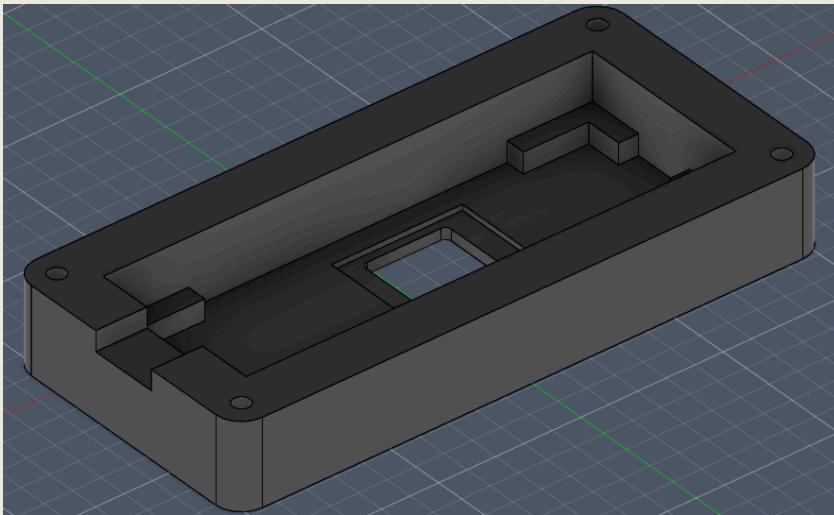
now I extrude the lid:



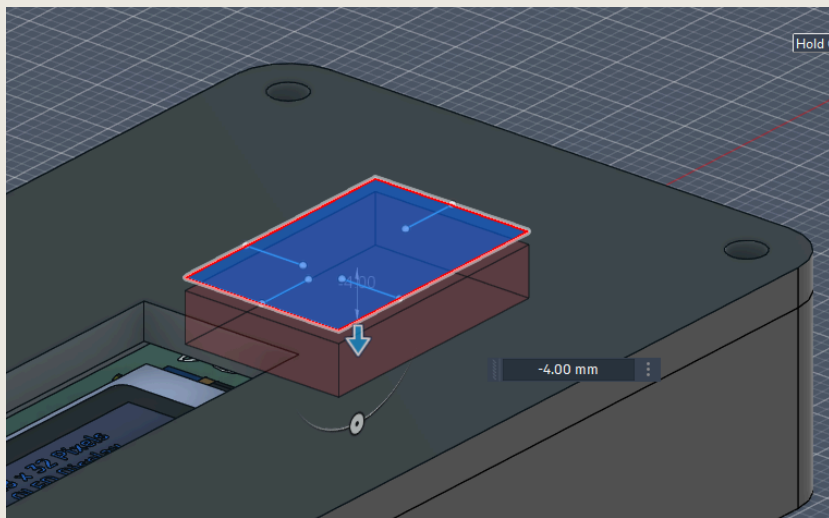
now I make another cutout for the usb c:



now I add these raisers for the pcb:

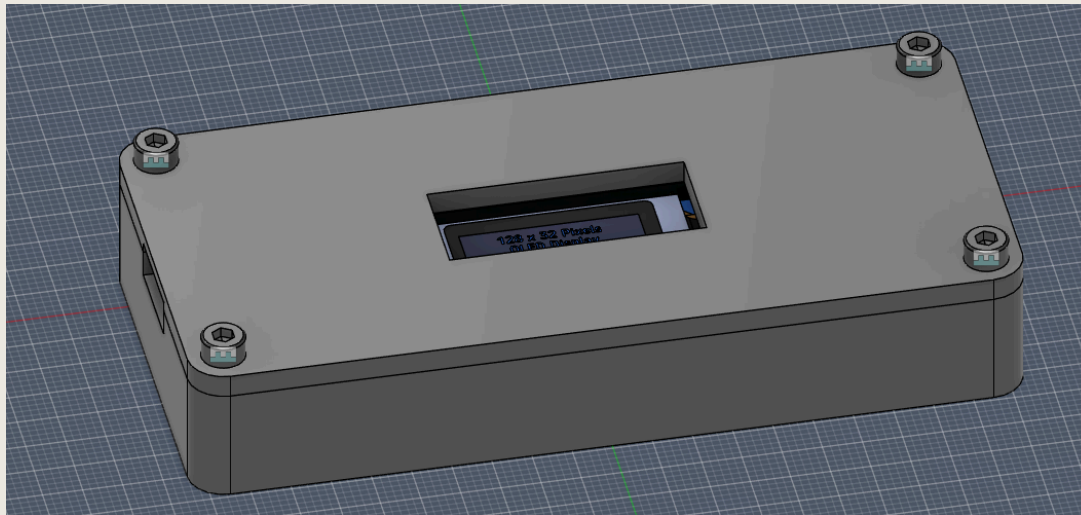


now I add this thin area to allow more light to come through:



now that area is only 1mm thick

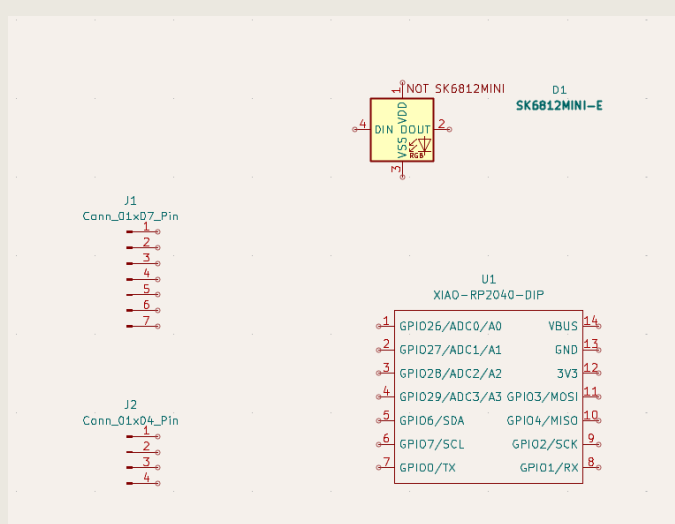
here is the final case:



Cubic-Vyper added to the journal **DESIGN** 2.75h logged

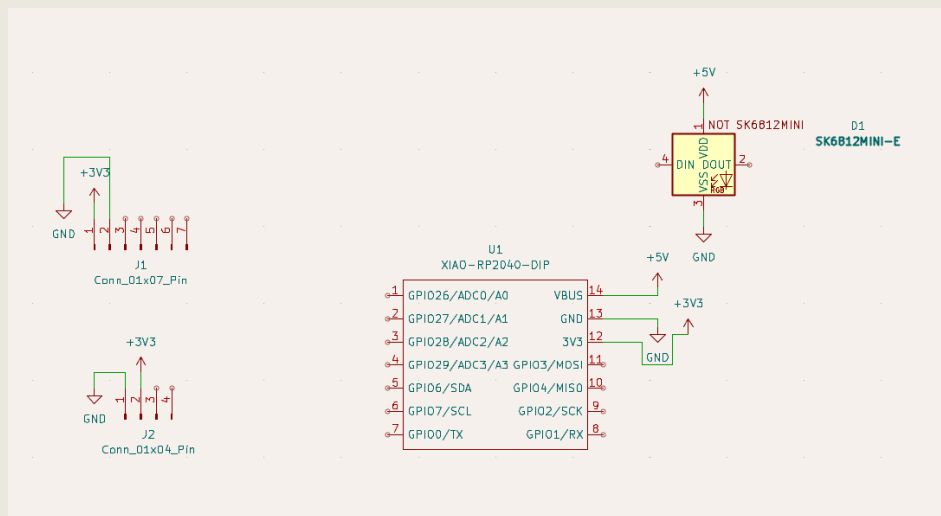
Mar 19, 2026

First I import all my parts into my schematic, this took a while, I didn't know that pin headers where called Conn_01x04_Pin,

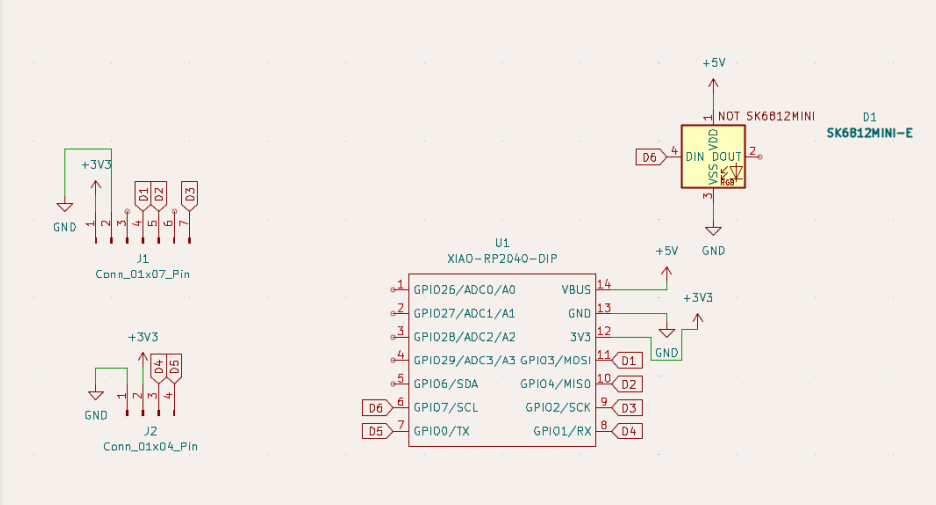


the 7pin header the the color sensor and the 4pin is the OLED

now I place all the power symbols



Now I wire all of my data pins:

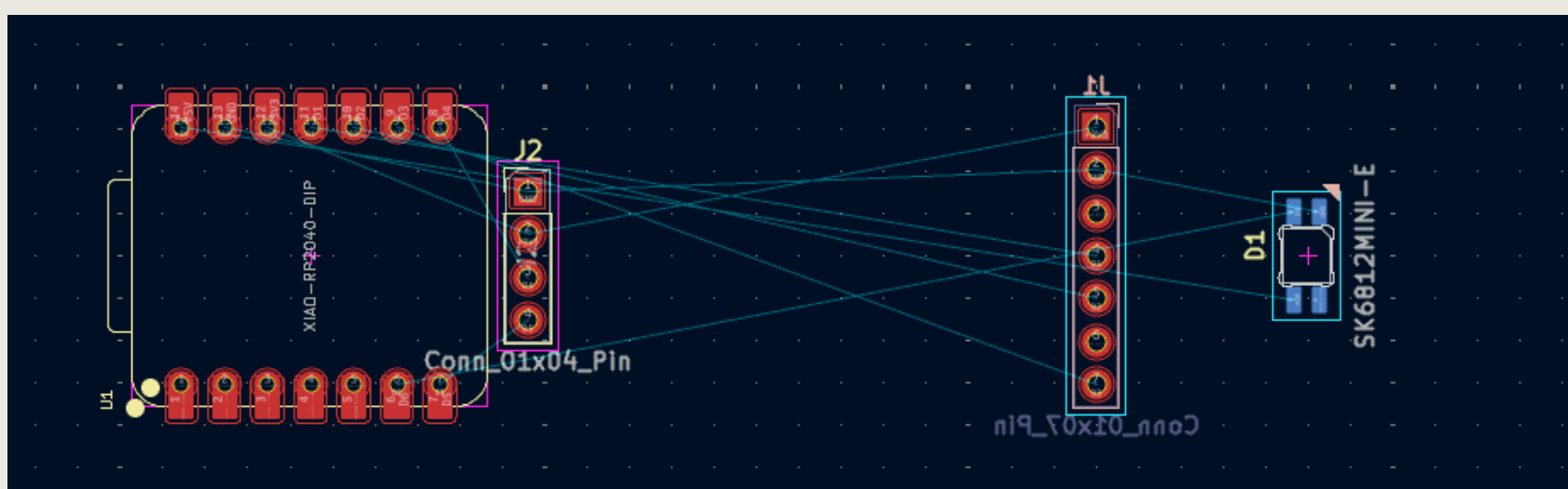


pin 3 (7pin header) is 3v3, i didn't wirre this because I already have it connected to VIN(pin 1), and pin 6 (7pin header) is INT, witch is unnneeded.

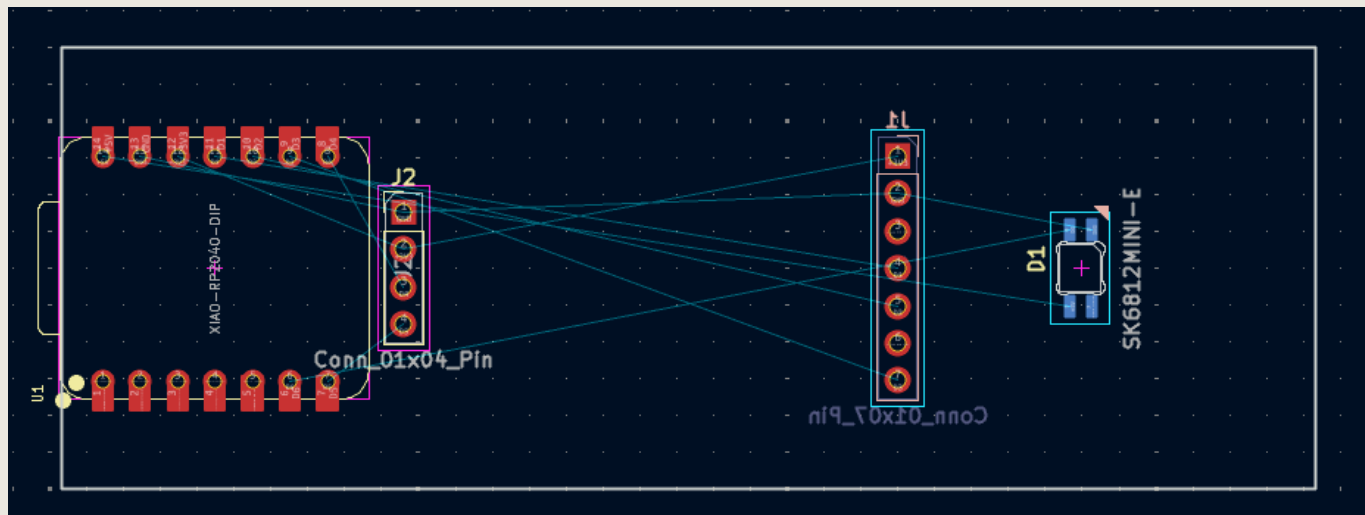
now I assign the footprints:

Symbol : Footprint Assignments			
1	D1 -	SK6812MINI-E :	Seeed Studio XIAO Series Library:SK6812MINI-E_fixed
2	J1 -	Conn_01x07_Pin :	Connector_PinHeader_2.54mm:PinHeader_1x07_P2.54mm_Ver
3	J2 -	Conn_01x04_Pin :	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Ver
4	U1 -	XIAO-RP2040-DIP :	Seeed Studio XIAO Series Library:XIAO-RP2040-DIP

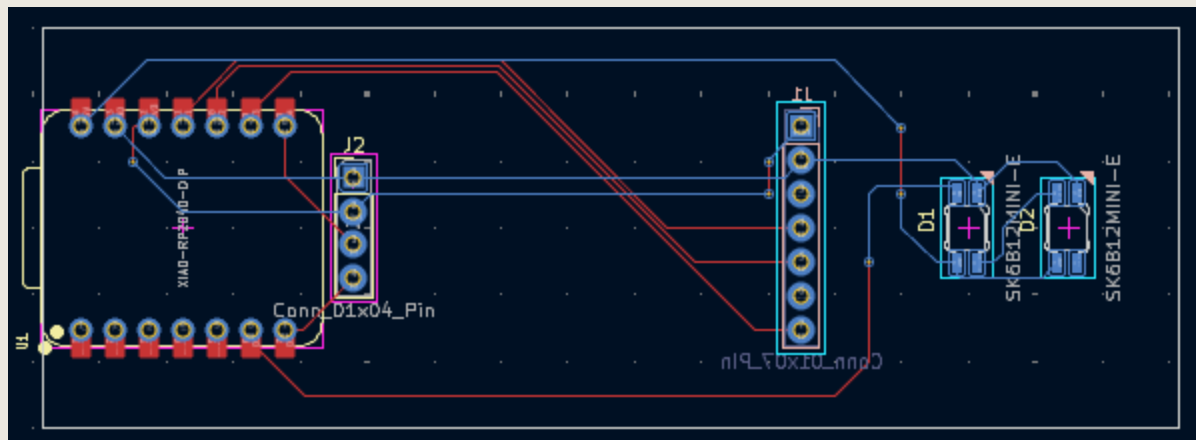
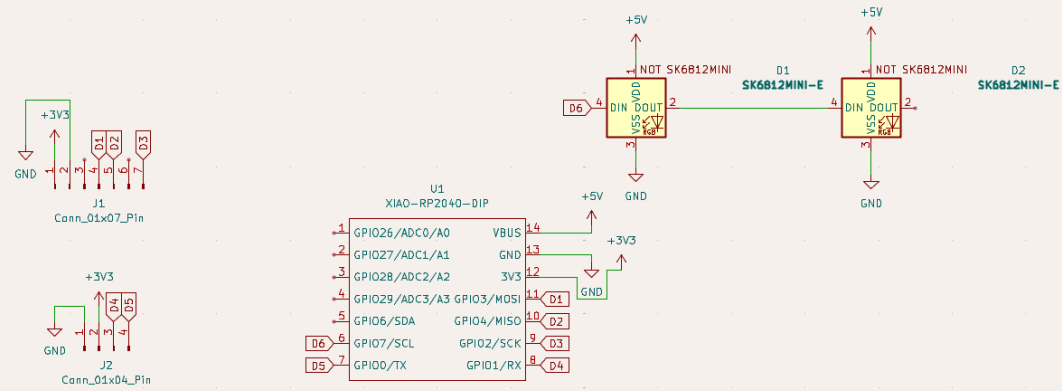
now I import the footprint into the pcb editor and move them around so everything will fit:



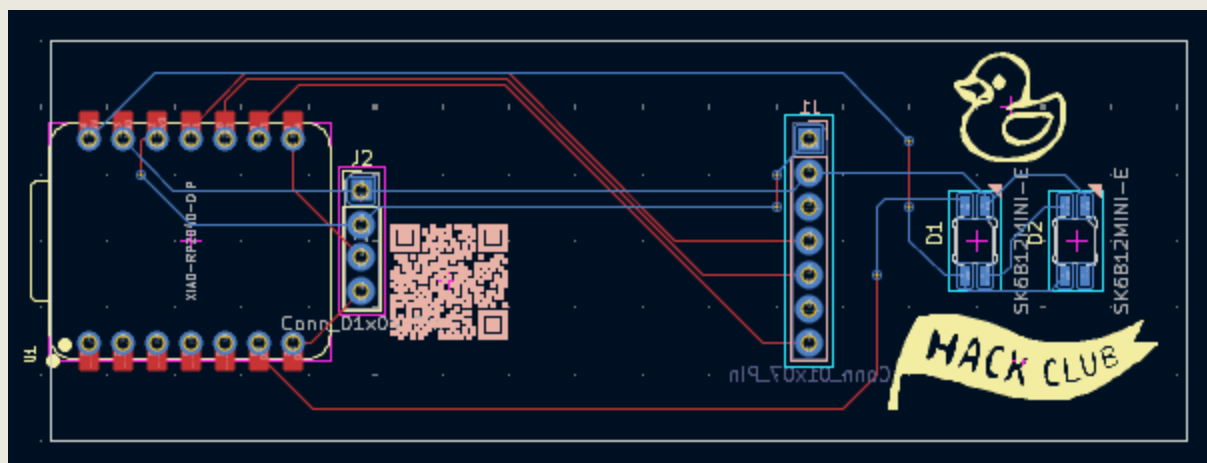
now I add the borrdor:



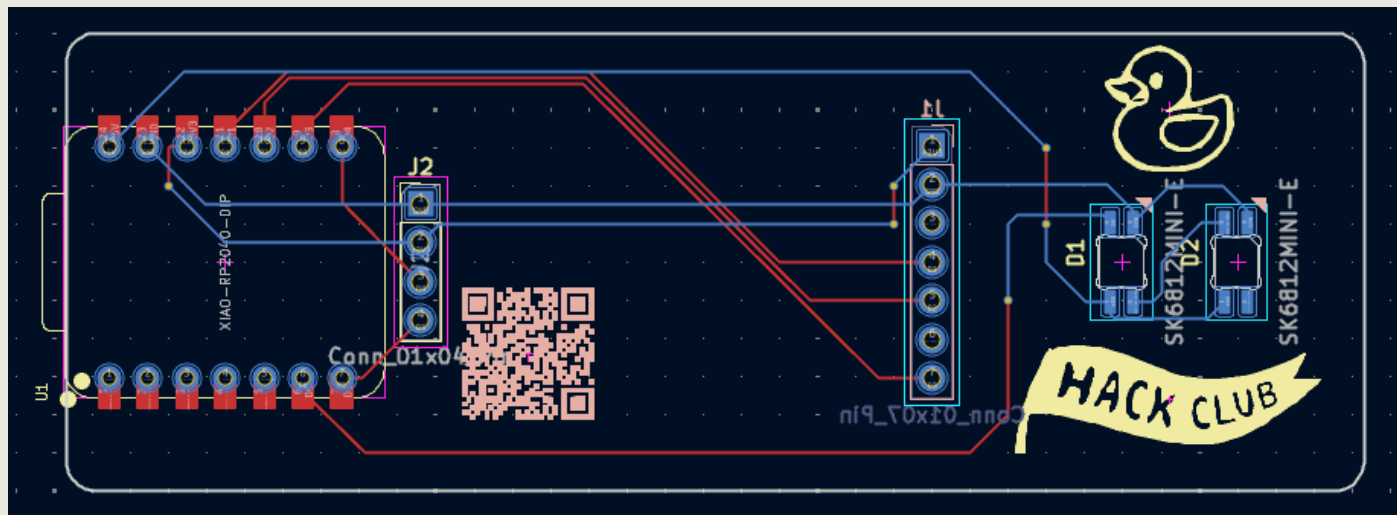
I decided to add another led:



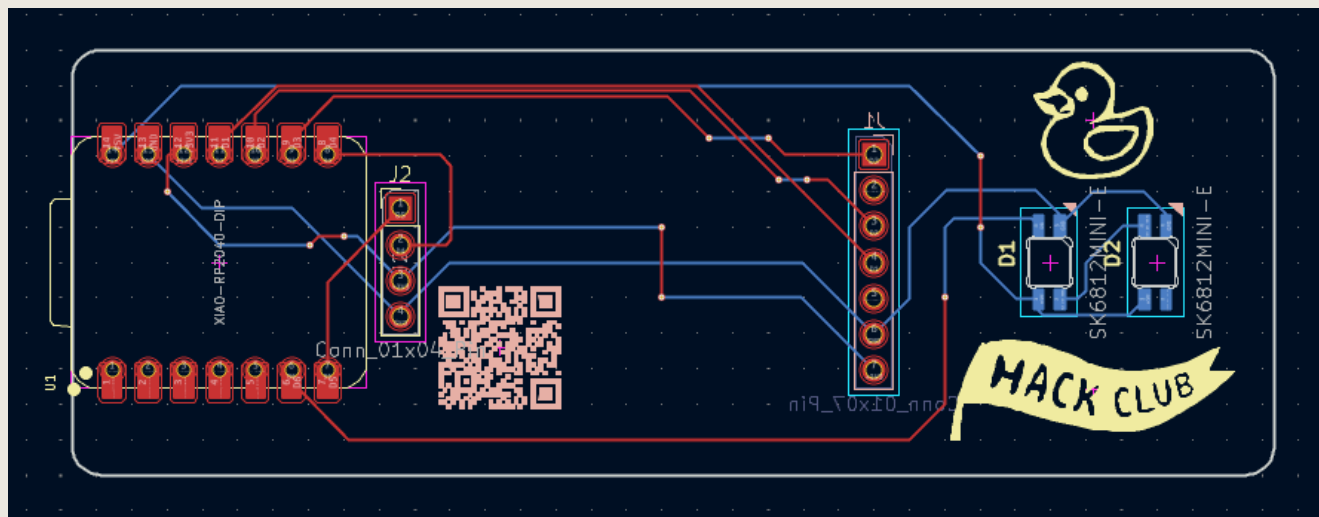
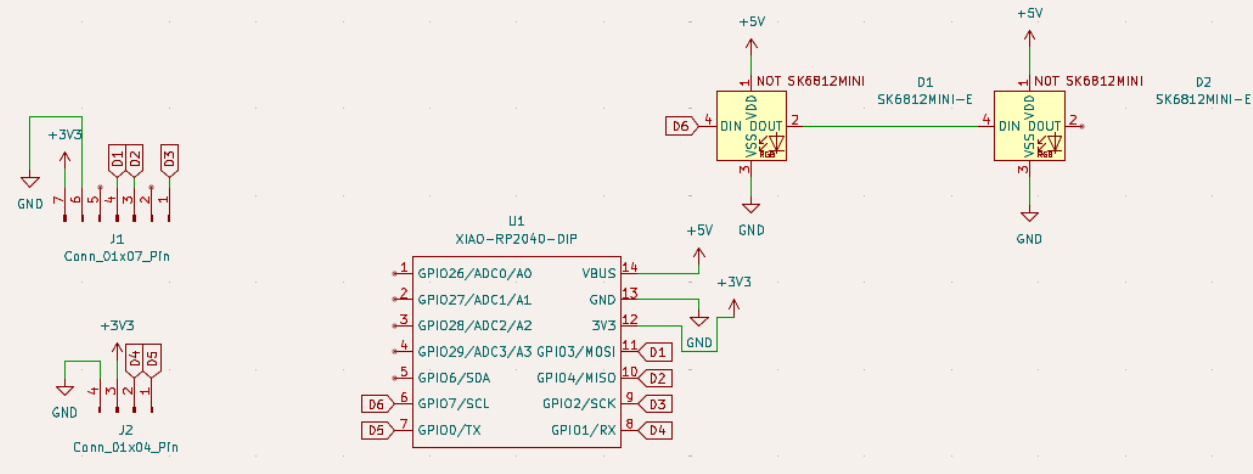
now the wiring is done, now i will add some graphics:



I also added round corners:

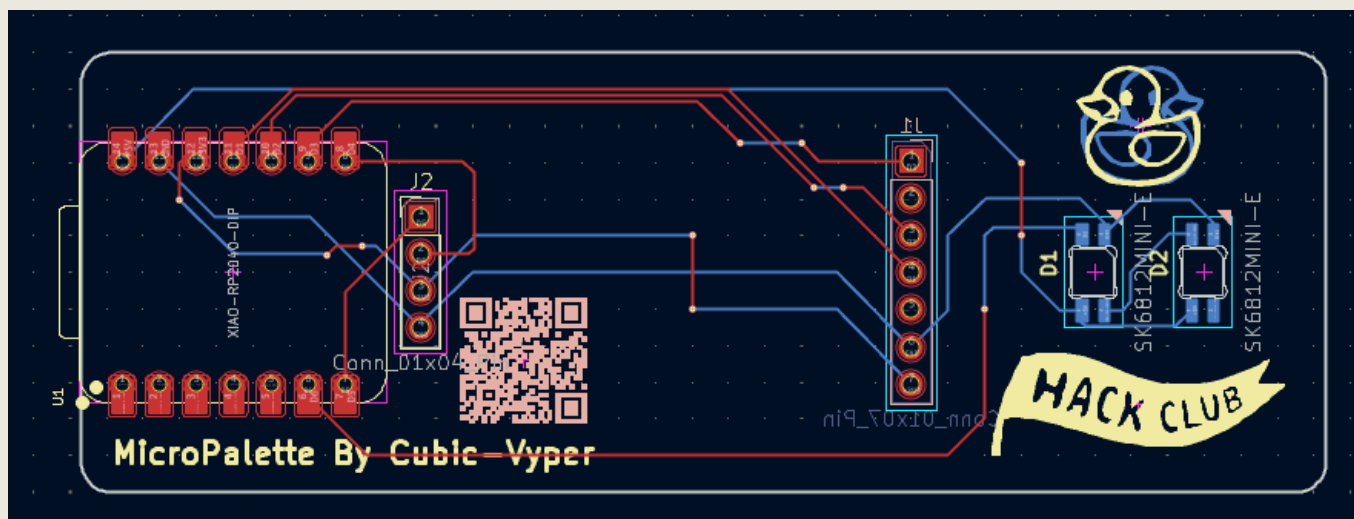


I just found out that I wired both pin headers backwards:



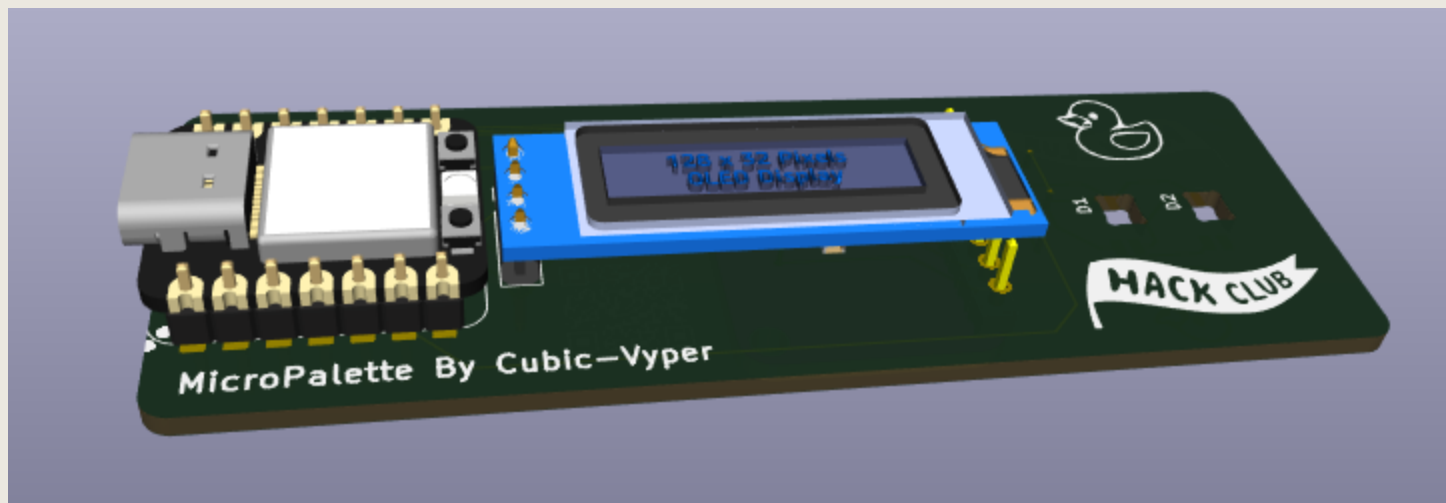
Fixed!

I decided to add a duck thats exposed like a soldering pad (B.Cu, B.Paste, B.Mask) to go on the back.

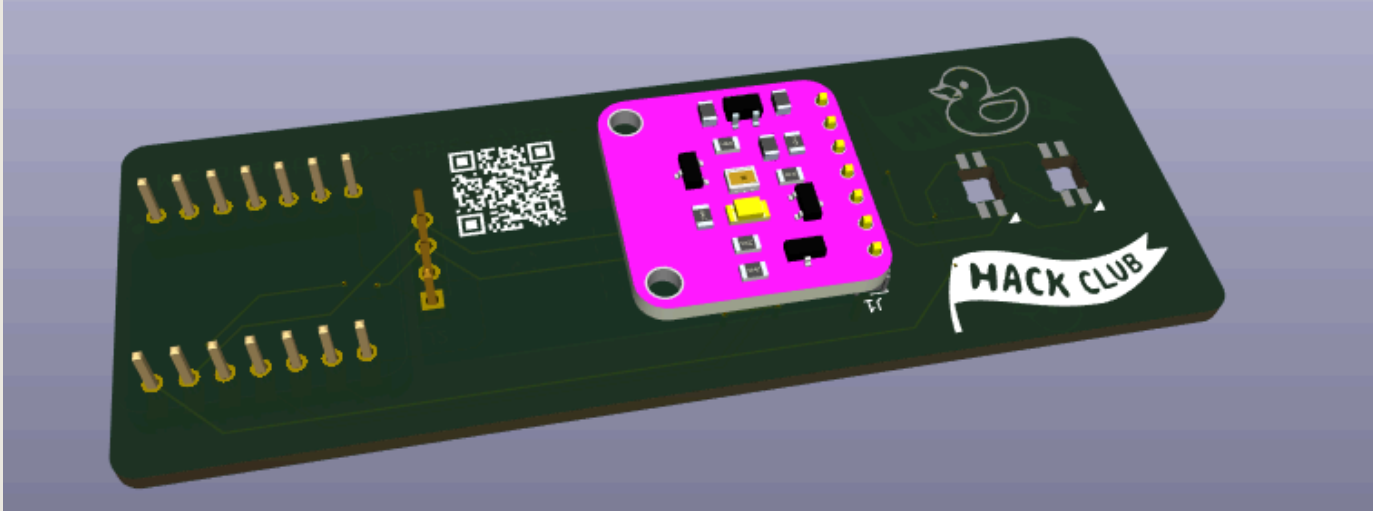


now i add all the 3d models:

Top:



Bottem:



I also decided to add another flag to the back I also flipped the duck:

